Udbhav Ram

41 Rotunda Street
Brampton, ON, Canada
(647) 339-8726
udbhavram41@gmail.com
sites.google.com/view/udbhav-ram
github.com/udiram
linkedin.com/in/udbhav-ram-engineering-and-medicine

Education

B.Sc. **Honours Medical Physics (with Co-op)** **Sept 2021–May 2026**
McMaster University, Hamilton, ON

- Dean's List (2021–Present), Current cGPA: 9.5
- Relevant coursework: Clinical Applications of Physics in Medicine (Medphys 6T03), Radiation Biology (Medphys 6U03), Radiation and Radioisotope Methodology I (Medphys 4RA3)

Research Experience

International Visiting Scholar, Department of Radiation Oncology **Sept 2021–Present**
University of Alabama at Birmingham, Birmingham, AL

- Evaluated Ethos 2.0 High-Fidelity Mode for multi-met, single-isocenter stereotactic radiotherapy using semi-automated comparative analysis
- Assessed inter-observer variability for Ethos adaptive accelerated partial breast irradiation plans
- Integrated Large Language Model agents into clinical workflows for radiation oncology quality assurance applications
- Improved TG-263 target name compliance using locally-hosted Large Language Models
- Investigated the impact of genetics on radiosensitivity using deep learning and machine learning approaches
- Assessed efficacy of state-of-the-art deep learning models in abdominal organ auto-segmentation
- Optimized dose delivery during fractionated radiotherapy using deep learning and simulation techniques
- Completed 400 hours of clinical shadowing in radiation oncology and medical physics

Undergraduate Thesis Student, Juravinski Cancer Centre **Sept 2024–May 2025**
Hamilton, ON

- Conducted multi-institution commissioning study comparing dosimetric properties of 6X and 10X flattening filter-free photon beams on a TrueBeam iTX Linac for SBRT applications using the Eclipse TPS (JCC-UAB collaboration)

Researcher, Department of Medical Physics **Aug 2022–Jan 2023**
University of Western Ontario, London, ON

- Built deep learning models to provide real-time analysis of optical scans, determining hemoglobin oxygenation within the brain

Clinical Research Fellow, Department of Nephrology **Sept 2021–Aug 2022**
St. Joseph's Hospital, Hamilton, ON

- Designed deep learning computer vision applications to provide augmented reality surgery tools for 5/6 nephrectomy mouse models

High School Research Fellow, Department of Physics and Astronomy **Sept 2019–Aug 2022**
McMaster University, Hamilton, ON

- Investigated the effects of resveratrol, caffeine, β-carotene, and EGCG on amyloid aggregation in synthetic brain membranes

Research Assistant, W Booth School of Engineering Practice and Technology **Jan 2023–June 2023**
McMaster University, Hamilton, ON

- Built and validated full-stack applications using JavaScript frameworks to improve accuracy in academic transcript generation

Professional & Work Experience

Data and Strategy Intern, McLaren Racing, Arrow McLaren IndyCar **June 2023–Aug 2023**
Indianapolis, IN

- Developed modelling pipelines, stochastic simulators, and deep learning computer vision applications to assist race, performance, and data engineers

Lead, Data Analytics and Strategy, MAC Formula Electric **Oct 2021–Jan 2024**
Hamilton, ON

- Interfaced with vehicle control systems to provide information to the driver on the dashboard
- Developed CAN communications protocols and controls systems integration

Full Stack Engineer, Syth-Med Biotechnology **2020–2021**

- Developed full-stack applications for biotechnology solutions

DevOps Technical Lead, WaaW Global Inc. **2020–2021**

- Led DevOps infrastructure and technical operations

Ambassador, Journal of Science Policy & Governance **2022–2023**

- Served as student ambassador promoting science policy engagement and governance initiatives

Teaching & Mentoring Experience

Undergraduate Teaching Assistant, Department of Physics and Astronomy **Sept 2024–May 2025**
McMaster University, Hamilton, ON

- Undergraduate teaching assistant for Introductory Mechanics (1E03) and Electricity and Magnetism (1E03)

Mentor, McMaster Undergraduate Physics Society **Sept 2024–May 2025**
McMaster University, Hamilton, ON

- Mentoring physics students to help navigate courses, internships and extra-curriculars

Mentor, McMaster Science Careers and Experience Centre **Sept 2024–May 2025**
McMaster University, Hamilton, ON

- Mentoring students in the medical and biological physics co-op stream to find placements and provide comprehensive document and interview preparation.

Head Yoga Instructor, McMaster University Athletics **May 2022–Present**
Hamilton, ON

- Head yoga instructor teaching cohorts of 50 students per term

Publications

Ram, Udbhav S., Duan, Jingwei, Cardan, Rex A., Robbins, Anna, Dobelbower, Christopher, Speers, Corey, Stanley, Courtney B., Stanley, Dennis N., Boggs, Drexell H., Lacy, Jessie, Keene, Kimberly, Wine, Madelyin, Soike, Michael, Viscariello, Natalie N., Lima, Ndipnu, Tyler, Melissa, Cardenas, Carlos E., Pogue, Joel A. Evaluation of inter-observer variability for Ethos adaptive accelerated partial breast irradiation plans. *International Journal of Radiation Oncology, Biology, Physics*. Submission in progress. **2025**

Ram, Udbhav S., Ajay Subramanian, Jeffrey Peacock, A 200-gene machine-learning model predicts tumour preclinical and clinical radiosensitivity across dose levels, *Nature communications*. Submitted **2025**

Ram, Udbhav S., Soomro, Asfia, Gao, Bo, Krepinsky, Joan C. An Augmented Reality setup to improve accuracy in the 5/6 nephrectomy surgical model. *Computers in Biology and Medicine*. Submitted **2025**

Ram, Udbhav S., Cardan, Rex A., Cardenas, Carlos E. Improving TG-263 target name compliance using locally-hosted Large Language Models *Physics in Radiation Oncology*. Submission in progress **2025**

Ram, Udbhav S., Popple, Richard A., Fiveash, John B., Heinzman, Katherine A., Kumar, Yogesh, Viscariello, Natalie N., Stanley, Courtney B., Stanley, Dennis N., Cardenas, Carlos E., Pogue, Joel A. Evaluation of Ethos 2.0 High-Fidelity Mode for Multi-Met, Single-Isocenter Stereotactic Radiotherapy: A Semi-Automated Comparative Analysis. *Journal of Applied Clinical Medical Physics*. 10.1002/acm2.70370 **2025**

Ram, Udbhav S., Pogue, Joel A., Soike, Michael, Pfister, Neil T., Jacob, Rojymon, Cardenas, Carlos E. Assessing quantitative performance and expert review of multiple deep learning-based frameworks for computed tomography-based abdominal organ auto-segmentation *Intelligent Oncology*. 1(2), 10.1016/j.intonc.2025.03.003 **2024**

Gastaldo, Isabella P., Himbert, Sebastian, **Ram, Udbhav S.**, Rheinstädter, Maikel C. The Effects of Resveratrol, Caffeine, β-Carotene, and Epigallocatechin Gallate (EGCG) on Amyloid-β_{25-35} Aggregation in Synthetic Brain Membranes. *Molecular Nutrition & Food Research*, 10.1002/mnfr.202000632 **2020**

Conference Presentations & Invited Talks

Ram, Udbhav S., Pogue, Joel A., Cardenas, Carlos E. Evaluation of Ethos 2.0 High-Fidelity Mode for Multi-Met, single-isocenter stereotactic radiotherapy: A semi-automated comparative analysis. Invited Poster, Society of Physics Students Congress. **October 2025**

Ram, Udbhav S., Udbhav S. Ram, Carlos E. Cardenas, Marcin Wierzbicki Dosimetric Comparison of 6X and 10X Flattening Filter-Free Photon Beams on a Truebeam Linac Using the Eclipse TPS, Therapy Poster, American Association of Physicists in Medicine. **July 2025**

Ram, Udbhav S., Carlos E. Cardenas, Rex A. Cardan Improving TG-263 Target Name Compliance Using Locally-Hosted Large Language Models. Blue Ribbon Poster, American Association of Physicists in Medicine. **July 2025**

Ram, Udbhav S., Pogue, Joel A., Cardenas, Carlos E. Head-to-Head Comparison of State-of-the-Art AutoML Segmentation Frameworks Using Publicly Available Data. Poster at American Association of Physicists in Medicine Annual Meeting. **July 2024**

Ram, Udbhav S., Pogue, Joel A., Cardenas, Carlos E. Evaluation of Ethos 2.0 High-Fidelity Mode for Multi-Met, single-isocenter stereotactic radiotherapy: A semi-automated comparative analysis. Poster at American Association of Physicists in Medicine Annual Meeting. **July 2024**

Ram, Udbhav S., et al. Invited Speaker. Improving TG-263 target name compliance using locally-hosted large language models, Canadian Organization of Medical Physicists Annual Scientific Meeting, **May 2024**

Ram, Udbhav S., Peacock, Jeffrey, Subramanian, Ajay. Optimizing dose delivery during fractionated radiotherapy. Overall Best Talk Winner at Canadian Undergraduate Physics Conference. **Sept 2024**

Ram, Udbhav S., Pogue, Joel A., Soike, Michael, Pfister, Neil T., Jacob, Rojymon, Cardenas, Carlos E. Assessing Efficacy of Multiple, State Of The Art, Deep Learning Models in Abdominal Organ Auto-Segmentation. ePoster at American Association of Physicists in Medicine Annual Meeting. **July 2023**

Ram, Udbhav S., Pogue, Joel A., Soike, Michael, Pfister, Neil T., Jacob, Rojymon, Cardenas, Carlos E. Abdominal Multi-Organ Segmentation using Deep learning to enable enhanced workflows for LMIC. Overall 2nd Place Talk Winner at Physics Undergraduate Conference. **May 2023**

Ram, Udbhav S., Pogue, Joel A., Soike, Michael, Pfister, Neil T., Jacob, Rojymon, Cardenas, Carlos E. Abdominal organs auto-segmentation to reduce patient-wait times in radiation oncology treatments. 3rd place, Division of Medical and Biological Physics at Canadian Undergraduate Physics Conference. **Sept 2022**

Ram, Udbhav S., Pogue, Joel A., Soike, Michael, Pfister, Neil T., Jacob, Rojymon, Cardenas, Carlos E. Abdominal Multi Organ Segmentation using Deep learning. Talk at Canadian Undergraduate Medical Physics Conference. **Aug 2022**

Ram, Udbhav S., Soomro, Asfia, Gao, Bo, Krepinsky, Joan C. An Augmented Reality setup to improve accuracy in the 5/6 nephrectomy surgical model. Finalist, Division of Medical and Biological Physics at Canadian Association of Physicists Conference. **May 2022**

Ram, Udbhav S., Himbert, Sebastian, Rheinstädter, Maikel C. Quantitative analysis of Gel electrophoresis images using cloud based, machine learning analysis. Talk at Canadian Association of Physicists Conference. **May 2021**

Honors & Awards

Blue Ribbon Poster, American Association of Physicists in Medicine **2025**

Finalist, Co-op Student of the Year (10/659 students), McMaster University **2025**

Global Experience Award Winner ($2,000 per co-op term abroad), McMaster University **2025**

Outstanding Poster Presentation (Full scholarship to 2025 SPS PhysCon), AAPM/SPS **2024**

Best Talk 1st Prize Winner ($500 + expenses-paid trip to CAP Congress), CUPC **2023**

McMaster Academic Entrance Scholarship ($3,000), Faculty of Science **2021**

Advanced Placement Scholar With Distinction (Avg. 4.5/5 on 6 AP exams), College Board **2021**

Appointments & Leadership

International Visiting Scholar, University of Alabama at Birmingham **May 2025–Present**

Student Ambassador, Mary Heersink Institute for Global Health, UAB **Jan 2025–Present**

President, Hindu YUVA, McMaster University **Jan 2025–Present**

Vice President, Cultural Affairs, Hindu YUVA, University of Alabama at Birmingham **Jan 2024–Present**

Vice President, Social, McMaster Undergraduate Physics Society **Jan 2024–Present**

Community Service & Volunteer Work

Volunteer, Humber River Hospital (Medical Imaging, Surgical Inpatient, Information Services) **Oct 2021–Jan 2024**
Toronto, ON

- Helped guide patients and visitors, and restocked medical supplies

Certifications & Licenses

ETHOS Adaptive Radiotherapy Course, University of Alabama at Birmingham **2025**

Stereotactic Radiosurgery Course, University of Alabama at Birmingham **2025**

Advanced Open Water Diver, Professional Association of Diving Instructors **2025**

Grade C Racing License, Federation of Motor Sports Clubs of India **2023**

Private Pilot's License (In Progress), Transport Canada **2023–Present**

Member, Equestrian Canada **2023–Present**

Royal Conservatory of Music, Level 8 Violin, Level 9 Piano **2019**

Certified Yoga Teacher (RYT-200), Yoga Alliance **2019**

French Language Certification (DELF-B2), République Française **2019**

Technical Skills

Programming & Software: Python, Machine Learning, Deep Learning, Computer Vision, Full Stack Engineering, DevOps, MIDAS, Pi Toolbox, SBG RaceWatch

Research Areas: Data Science, Clinical Research, Molecular Dynamics, Radiation Oncology, Medical Physics, Genetics, Modelling, LLM Integration

Engineering: Motorsports Engineering, Race Engineering, Performance Engineering, AI, AR/MR/VR

Languages: English (Native), Tamil (Fluent), French (Fluent), Spanish (Intermediate)

Open-Source Contributions

Beta tester and contributor, OpenPilot (comma.ai)

Contributor, Project MONAI, (Project MONAI)

Contributor, Devin, (Devin.ai)